POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a GPU and in what ways its going to assist the CPU to enhance the computer overall performance? Explain video controller with frame buffer organization. 7

   b) A monitor has 1280×1024 resolution and refreshes at 75 fps. How many pixels could be accessed per second and what is the access time per pixel of the system. 8

2. a) What are the limitations of DDA? Write the algorithm (any case) which overcomes these limitations. 7

   b) Derive the ellipse drawing algorithm for both region with initial decision parameter. 8

3. a) Why we need homogeneous coordinate system in transformation? Prove that, two successive translations are additive and two successive scaling are multiplicative in nature by using homogeneous coordinate system. 8

   OR

   Define window port and view port. Derive the window to viewport transformation matrix.

   b) You are provided with the clipping rectangle with coordinates A(10,10) B(10,20) C(20,10) and D(20,20). Clip a line with endpoints P(5,15) and Q(25,15) with the clipping rectangle using Cohen-Sutherland Line clipping algorithm. 7

4. a) How the rotation is performed in 3D? Write the rotational matrix in homogeneous form for X-axis, Y-axis and Z-axis Rotation. 7

   b) What is the importance of curve in Computer Graphics? Explain Cubic Bezier curve with necessary derivations. 8

Notebook

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The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define computer graphics. Explain the use of Computer Graphics in creating realistic Virtual Reality based applications. 7

   b) What do you mean by hardware acceleration? Differentiate between Raster and Random scan display with its architectural diagram 8

2. a) In Case two raster systems with resolutions of 640 by 480 and 1024 by 600, how many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 75 frames per second? What is the access time per pixel in each system? 8

   b) What role does bits in a frame buffer play in glowing the pixels on the screen with different intensities? Explain with an example. 7

3. a) Derive the algorithm for digitizing a circle with a radius of 'r' pixels with the starting pixel located at (0,r) and moving in an anticlockwise direction. 8

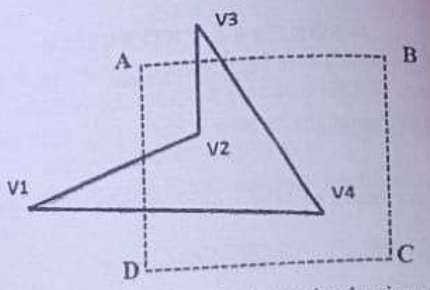
   b) What is scan conversion? Write the Bresenham's line drawing algorithm for slope less than 1 and derive its initial decision parameter. 7

4. a) What will be the final coordinates of a polygon with vertices A(3,4) B(5,4) C(5,2) D(3,4) after it is reflected about a line y=2x+1? 8

   b) Use Cohen-Sutherland algorithm to clip the line P1 (70, 20) and P2 (100, 10) against the window (50, 10) and (80,40). 7

OR

Clip the given polygon by using Sutherland-Hodgeman polygon clipping algorithm.



5. a) How can the visibility of objects be determined using object space method? Explain its advantages over image space methods. 8

   b) Explain the process of clipping 3D objects and mention how it differs from 2D clipping. 7

6. a) Explain Depth Buffer algorithm for visible surface detection. Also list its pros and cons. 8

   b) Explain constant shading method and mention its limitation. Propose and explain the method to overcome the limitation of constant shading. 7

7. Write short notes on: **(Any two)** 2×5

   a) Graphical language

   b) Viewing pipeline

   c) Beizer curve

POKHARA UNIVERSITY

Semester: Fall

Level: Bachelor
Programme: BE
Course: Computer Graphics (New)

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What do you mean by Computer Graphics? Explain the working principle of refresh CRT with figure. 7

   b) What are the benefits of using a GPU for graphical rendering? What makes a GPU more ideal for graphical rendering than using a CPU? 8

2. a) Mention the limitations of DDA algorithm. Derive the necessary expression of decision parameters to rasterize a line with m<1. 8

   b) Digitize one octant of a circle by using midpoint circle generation algorithm center at origin and radius is 12. 7

3. a) A graphics designer needs to animate the swing of pendulum in center of display window with string attached at pivot point (350,300). If pendulum is displaced by 45 degrees from its resting position. Calculate the extreme points of bob's trajectory if string length is of 10 units. 8

OR

Reflect the triangle A(4,3), B(5,5) and C(4,6) about an axis defined by equation of line y=2-2x.

   b) Why is it required to map an object description from a window to a viewport? A triangle with vertices A(30,30) B(50,30) C(40,35) is required to be mapped from a window with its lower left corner at (20,20) and upper right corner at (60,40) to a viewport with its lower left corner at (50,50) and upper right corner at (80,70). 7

4. a) Derive composite matrix for rotation about an arbitrary axis in 3D. 8

OR

What are non-planar surface? How can a Bezier surface be generated from Bezier curve?

   b) Rotate the triangle with vertices A(3,3) , B(5,3) , C(4,5) by 45° in anti clockwise direction While keeping B (3,3) fixed. 7

5. a) What role does Visible Surface detection play in displaying realistic scenes? Explain Depth Sorting approach for removing hidden surfaces. 8

b) What is Gouraud shading? Explain it with an example. What are its drawbacks? How such drawbacks are removed? 7

6. a) What is OpenGL? Write any five GLUT call back functions with their necessary syntax. 7

b) Why is it required to visualize data sets? Explain how data sets can be visualized? 8

7. Write short notes on: (**Any two**) 2×5

a) 2D Viewing pipeline

b) Perspective Projection

c) Window to viewport transformation

OR

Differentiate between parallel and perspective projection with real life example for each projection. 7

5. a) Define visible surface detection algorithm. Explain accumulation buffer method for visible surface detection. 8

   b) What is illumination model? Explain specular reflection and Phong model. 7

6. a) What is the significance of machine independent graphical language? Explain GKS and PHIGS. 8

   b) Draw the architectural diagram of OpenGL and explain different callback function. 2×5

7. Write short notes on: **(Any two)**

   a) 2D viewing pipeline

   b) Fractal Geometry method

   c) Color models

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